BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 203 · Karnaugh maps and simplification

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Simplify ¬(A ∨ B).

Q2. Keys 7 and 12 hash to index 2 in a table. What is that called?

Q3. What is 1 XOR 1?

Q4. If A changes from 0 to 1 while B is always 1, which variable is constant?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Karnaugh map — Grid for simplifying Boolean functions using adjacent combinations.

Gray code — Ordering in which adjacent binary labels differ by one bit.

group — A rectangle containing 1, 2, 4, 8, … adjacent 1-cells.

constant variable — A variable with the same value in all cells of a group.

wraparound — The opposite edges of a Karnaugh map are adjacent.

New learning

Read the labels — Rows represent A, and columns use Gray order BC: 00, 01, 11, 10.

Group neighbouring ones — The four 1s form one 2×2 rectangle. Adjacency across rows is allowed.

Keep what is constant — A and C change through the group, but B remains 0. So the simplified result is ¬B.

Worked example

Four 1s fill the entire A=0 row. Simplify the output.

1. Predict and explain — B and C each vary across the row; neither is needed in the expression.

2. Check and extend — A stays 0 in the whole group. The simplified term is ¬A.

We Do · Practise with guidance

Four 1s fill columns BC=11 and BC=10 across both rows. Simplify.

```
BC columns: 00 01 11 10
A=0:        0  0  1  1
A=1:        0  0  1  1
```

Predict and explain

A changes between rows, and C changes between columns 11 and 10.

Check and extend

B remains 1, so the simplified term is B.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Why do we group neighbouring 1s in a Karnaugh map?

Question 2

The whole A=1 row contains four 1s. Which simplified term remains?

Question 3

Columns BC=00 and BC=10 are joined by wraparound adjacency. Four 1s occupy both these columns. Simplify.

Exit Ticket · Answer before checking

A group of four 1s can remove two changing variables.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. ¬A ∧ ¬B.

Q2. A collision.

Q3. 0.

Q4. B, at value 1.

You Do pages · Independent questions

Q1. To simplify the Boolean expression by removing variables that change inside a group.

Q2. A. Both B and C change within the row, while A remains 1.

Q3. ¬C. A changes between the rows and B changes between 00 and 10; C stays 0.

Exit Ticket · Model answer

True. The group leaves only variables constant across all four cells.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: